ANLP Proofs

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1 Mathematical Motivation of Activation Steering Methods

Let

$$h_\ell \in \mathbb{R}^d$$

denote the hidden representation of an input at transformer layer ℓ. The objective of activation steering is to modify h_ℓ at inference time such that the resulting model exhibits a desired behavior while preserving as much of the original representation as possible.

We consider four increasingly structured steering mechanisms:

1. additive activation steering,
2. covariance-aware conceptor steering,
3. residual conceptor steering, and
4. mean-corrected residual conceptor steering.

The progression can be viewed as successively relaxing the assumptions of the preceding method:

$$\boxed{\begin{aligned}&\text{Single direction} \rightarrow \text{Structured subspace}\\ &\qquad\qquad\rightarrow \text{Controlled transformation}\\ &\qquad\qquad\rightarrow \text{Mean-preserving transformation}\end{aligned}} \tag{1}$$

1.1 Additive Activation Steering

Motivation. The simplest assumption is that a desired behavior can be represented by a single direction in activation space. If $v \in \mathbb{R}^d$ denotes this direction, steering is performed by

$$\boxed{h'_\ell = h_\ell + \beta v,} \tag{2}$$

where $\beta \in \mathbb{R}$ controls the strength and direction of the intervention.

This formulation is attractive because it makes the smallest possible structural assumption: the desired behavioral change can be approximated by a translation along one direction.

For emotion or attribute steering, a common construction is a difference between the mean activations of a target condition and a reference condition. Let

$$\mathcal{X}^+ = \{x_1^+, \ldots, x_n^+\}$$

denote activations collected under a target condition and

$$\mathcal{X}^- = \{x_1^-, \ldots, x_n^-\}$$

denote activations collected under a reference condition.

Define their empirical means as

$$\mu^+ = \frac{1}{n}\sum_{i=1}^{n} x_i^+, \qquad \mu^- = \frac{1}{n}\sum_{i=1}^{n} x_i^-. \tag{3}$$

The corresponding mean-difference steering vector is

$$\boxed{v = \mu^+ - \mu^-.} \tag{4}$$

Why does the mean difference arise naturally? The mean-difference vector can be justified as the optimal constant translation that aligns the reference activation distribution with the target distribution.

Consider paired activations x_i^- and x_i^+, where x_i^- is obtained from the reference condition and x_i^+ is obtained from the target condition.

We seek a single vector v that is added to every reference activation such that the resulting activation is as close as possible to its corresponding target activation. This can be formulated as the following least-squares problem:

$$\boxed{\min_v \frac{1}{n}\sum_{i=1}^{n} \left\|(x_i^- + v) - x_i^+\right\|_2^2} \tag{5}$$

In other words, the objective asks:

> What single vector v should be added to every reference activation so that the resulting activations are, on average, as close as possible to the corresponding target activations?

Define the paired activation difference as

$$\Delta_i = x_i^+ - x_i^-. \tag{6}$$

Then,

$$(x_i^- + v) - x_i^+ = v - (x_i^+ - x_i^-) \tag{7}$$

$$= v - \Delta_i. \tag{8}$$

Therefore, Equation (5) can be rewritten as

$$\boxed{\min_{v} \frac{1}{n} \sum_{i=1}^{n} \|v - \Delta_i\|_2^2 .} \tag{9}$$

This objective asks us to find the single vector v that is closest, in the least-squares sense, to all the individual activation differences Δ_i.

Differentiating Equation (9) with respect to v gives

$$\nabla_v \left[\frac{1}{n} \sum_{i=1}^{n} \|v - \Delta_i\|_2^2 \right] = \frac{2}{n} \sum_{i=1}^{n} (v - \Delta_i). \tag{10}$$

At the optimum, the gradient must be zero. Hence,

$$\frac{2}{n} \sum_{i=1}^{n} (v - \Delta_i) = 0, \tag{11}$$

$$\sum_{i=1}^{n} (v - \Delta_i) = 0, \tag{12}$$

$$\sum_{i=1}^{n} v = \sum_{i=1}^{n} \Delta_i. \tag{13}$$

Since v is the same vector for every example,

$$nv = \sum_{i=1}^{n} \Delta_i, \tag{14}$$

$$v = \frac{1}{n} \sum_{i=1}^{n} \Delta_i. \tag{15}$$

Substituting the definition of Δ_i from Equation (6),

$$v = \frac{1}{n} \sum_{i=1}^{n} \left(x_i^+ - x_i^- \right) \tag{16}$$

$$= \frac{1}{n} \sum_{i=1}^{n} x_i^+ - \frac{1}{n} \sum_{i=1}^{n} x_i^- . \tag{17}$$

Defining the empirical means

$$\mu^+ = \frac{1}{n} \sum_{i=1}^{n} x_i^+, \qquad \mu^- = \frac{1}{n} \sum_{i=1}^{n} x_i^-, \tag{18}$$

we finally obtain

$$\boxed{v = \mu^+ - \mu^- .} \tag{19}$$

Therefore, the commonly used mean-difference steering vector is not merely a heuristic. Under the squared-error objective in Equation (5), it is the *least-squares optimal single translation* between the paired reference and target activation distributions.

This provides a mathematical justification for additive steering: when the desired behavioral change can be adequately represented by a single direction, the mean-difference vector is the optimal constant translation under this objective.

1.2 Covariance-Aware Conceptor Steering

Motivation. The limitation of additive steering motivates a richer question:

> What if the desired behavior is represented not by a single direction, but by a structured region or subspace of activation space?

Instead of representing the target behavior using only a mean difference, let

$$X \in \mathbb{R}^{n \times d}$$

contain n activation vectors associated with the desired behavior.

Define the empirical correlation matrix

$$\boxed{R = \frac{X^\top X}{n}.} \tag{20}$$

The conceptor matrix is obtained by finding a linear transformation that reconstructs the observed activation patterns while penalizing an unnecessarily large transformation:

$$\boxed{\min_C \ \|X - XC\|_F^2 + \alpha^{-2}\|C\|_F^2.} \tag{21}$$

The first term,

$$\|X - XC\|_F^2,$$

encourages C to preserve the activation structure.

The second term,

$$\alpha^{-2}\|C\|_F^2,$$

prevents the solution from simply becoming an unrestricted reconstruction operator. The aperture α controls the strength of this regularization.

Without the regularizer, $C = I$ would obtain zero reconstruction error:

$$\|X - XI\|_F^2 = 0.$$

Such a solution would not perform meaningful filtering.

1.2.1 Derivation of the conceptor matrix

Expanding the Frobenius norm gives

$$\mathcal{L}(C) = \operatorname{tr}\left[(X - XC)^\top (X - XC)\right] + \alpha^{-2} \operatorname{tr}(C^\top C) \tag{22}$$

$$= \operatorname{tr}(X^\top X) - 2\operatorname{tr}(CX^\top X) + \operatorname{tr}(C^\top X^\top XC) + \alpha^{-2} \operatorname{tr}(C^\top C). \tag{23}$$

Using

$$X^\top X = nR,$$

and ignoring the positive multiplicative constant n, the objective can be expressed in terms of R.

The gradient with respect to C is

$$\nabla_C \mathcal{L} = 2CR - 2R + 2\alpha^{-2}C. \tag{24}$$

At the optimum,

$$\nabla_C \mathcal{L} = 0,$$

and therefore

$$CR - R + \alpha^{-2}C = 0, \tag{25}$$

$$C(R + \alpha^{-2}I) = R. \tag{26}$$

Since

$$R + \alpha^{-2}I \succ 0,$$

it is invertible. Hence the unique solution is

$$\boxed{C = R(R + \alpha^{-2}I)^{-1}.} \tag{27}$$

Thus, the conceptor is not an arbitrary projection matrix. It is the optimal regularized linear reconstruction operator for the observed activation distribution.

1.2.2 Spectral interpretation

Let

$$R = U\Lambda U^\top$$

be the eigendecomposition of R, where

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d).$$

Substituting this decomposition into Equation (27) gives

$$C = U \operatorname{diag}\left(\frac{\lambda_i}{\lambda_i + \alpha^{-2}}\right)_{i=1}^{d} U^\top. \tag{28}$$

Hence, for every principal direction u_i,

$$\boxed{Cu_i = \gamma_i u_i, \qquad \gamma_i = \frac{\lambda_i}{\lambda_i + \alpha^{-2}}.} \tag{29}$$

Therefore,

$$0 \le \gamma_i < 1.$$

A direction with large variance satisfies

$$\lambda_i \gg \alpha^{-2} \quad \Longrightarrow \quad \gamma_i \approx 1,$$

whereas a direction with small variance satisfies

$$\lambda_i \ll \alpha^{-2} \quad \Longrightarrow \quad \gamma_i \approx 0.$$

Thus, the conceptor implements a *soft projection*:

$$\boxed{\text{high-variance directions are preserved,} \qquad \text{low-variance directions are attenuated.}} \tag{30}$$

Unlike a hard projection, whose eigenvalues are restricted to 0 or 1, the conceptor permits continuous attenuation.

1.2.3 Conceptor-based steering

The original conceptor steering operation is

$$\boxed{h'_\ell = \beta C_\ell h_\ell.} \tag{31}$$

In the eigenbasis of C, write

$$h_\ell = \sum_{i=1}^{d} a_i u_i.$$

Then

$$h'_\ell = \beta \sum_{i=1}^{d} \gamma_i a_i u_i. \tag{32}$$

Consequently, the transformation is input-dependent and direction-dependent:

$$\boxed{a'_i = \beta \frac{\lambda_i}{\lambda_i + \alpha^{-2}} a_i.} \tag{33}$$

This is fundamentally different from additive steering. Additive steering translates every input by the same vector v, whereas conceptor steering transforms each principal direction according to the learned geometry of the activation distribution.

Limitation. The additional expressivity comes at a cost. If a simple task is effectively represented by one dominant direction, a full covariance-aware transformation may modify directions that are not needed for the task. Hence, conceptor steering should not be assumed to dominate additive steering for low-dimensional tasks.

1.3 Residual Conceptor Steering

Motivation. The published conceptor operation

$$h'_\ell = \beta C_\ell h_\ell$$

completely replaces the original activation.

This creates a potentially undesirable behavior: even when the original hidden representation already contains useful contextual or semantic information, the conceptor transformation modifies every component of the representation.

We therefore introduce a residual coefficient

$$\lambda \in [0, 1]$$

and define

$$\boxed{h'_\ell = h_\ell + \lambda(\beta C_\ell - I)h_\ell.} \tag{34}$$

Rearranging,

$$h'_\ell = h_\ell + \lambda\beta C_\ell h_\ell - \lambda h_\ell \tag{35}$$

$$= (1 - \lambda)h_\ell + \lambda\beta C_\ell h_\ell. \tag{36}$$

Therefore,

$$\boxed{h'_\ell = (1 - \lambda)h_\ell + \lambda(\beta C_\ell h_\ell).} \tag{37}$$

This reveals the role of λ directly.

For

$$\lambda = 0,$$

we obtain

$$h'_\ell = h_\ell,$$

corresponding to the unsteered model.

For

$$\lambda = 1,$$

we obtain

$$h'_\ell = \beta C_\ell h_\ell,$$

which exactly recovers the original conceptor operation.

Thus, the residual formulation is a strict generalization of the published conceptor method.

1.3.1 Bound on the intervention

The change in the hidden state is

$$h'_\ell - h_\ell = \lambda(\beta C_\ell - I)h_\ell. \tag{38}$$

Taking the Euclidean norm,

$$\|h'_\ell - h_\ell\|_2 \leq \lambda \|\beta C_\ell - I\|_2 \|h_\ell\|_2. \tag{39}$$

Hence,

$$\boxed{\|h'_\ell - h_\ell\|_2 \propto \lambda} \tag{40}$$

for fixed C_ℓ, β, and h_ℓ.

This provides a mathematical motivation for residual steering: λ directly controls the maximum magnitude of the intervention relative to the original representation.

In particular, reducing λ prevents the conceptor from completely replacing the model's original hidden state.

1.3.2 Spectral interpretation

For a principal direction u_i satisfying

$$C_\ell u_i = \gamma_i u_i,$$

the residual transformation becomes

$$h'_\ell = \left[(1-\lambda)I + \lambda\beta C_\ell\right] h_\ell. \tag{41}$$

Therefore, along u_i,

$$\boxed{a'_i = \left[(1-\lambda) + \lambda\beta\gamma_i\right] a_i.} \tag{42}$$

Since

$$\gamma_i = \frac{\lambda_i}{\lambda_i + \alpha^{-2}},$$

we obtain

$$\boxed{a'_i = \left[(1-\lambda) + \lambda\beta\frac{\lambda_i}{\lambda_i + \alpha^{-2}}\right] a_i.} \tag{43}$$

This makes the roles of the three parameters explicit:

$$\boxed{\alpha \rightarrow \text{selectivity of the learned subspace,}} \tag{44}$$

$$\boxed{\beta \rightarrow \text{amplification of selected directions,}} \tag{45}$$

$$\boxed{\lambda \rightarrow \text{strength of the intervention relative to the base state.}} \tag{46}$$

1.4 Mean-Corrected Residual Conceptor

Motivation. Residualization controls the magnitude of the intervention, but it does not by itself prevent a conceptor from changing the mean of the activation distribution.

Let

$$\mu_\ell = \mathbb{E}[h_\ell].$$

Consider first the simple conceptor transformation

$$h'_\ell = C_\ell h_\ell.$$

Taking expectations gives

$$\mathbb{E}[h'_\ell] = C_\ell \mathbb{E}[h_\ell] = C_\ell \mu_\ell. \tag{47}$$

In general,

$$C_\ell \mu_\ell \neq \mu_\ell.$$

Therefore,

$$\boxed{\mathbb{E}[h'_\ell] \neq \mu_\ell.} \tag{48}$$

This means that direct conceptor projection can shift the centroid of the activation distribution.

For activation steering, this is important because the centroid may contain useful contextual or attribute information. The objective is therefore to transform the variation around the centroid while preserving the centroid itself.

1.4.1 Derivation of the bias-recovery term

Suppose we use an affine transformation

$$h'_\ell = C_\ell h_\ell + b_\ell. \tag{49}$$

We require mean preservation:

$$\mathbb{E}[h'_\ell] = \mu_\ell. \tag{50}$$

Taking expectations,

$$\mathbb{E}[h'_\ell] = C_\ell \mu_\ell + b_\ell. \tag{51}$$

Therefore, the required bias is obtained by solving

$$C_\ell \mu_\ell + b_\ell = \mu_\ell,$$

which gives

$$\boxed{b_\ell = \mu_\ell - C_\ell \mu_\ell = (I - C_\ell)\mu_\ell.} \tag{52}$$

When the full conceptor transformation includes the scaling factor β,

$$h'_\ell = \beta C_\ell h_\ell + b_\ell,$$

the same argument gives

$$\beta C_\ell \mu_\ell + b_\ell = \mu_\ell, \tag{53}$$

and hence

$$\boxed{b_\ell = \mu_\ell - \beta C_\ell \mu_\ell.} \tag{54}$$

This is the bias-recovery term used by the mean-corrected conceptor method.

1.4.2 Mean-centering interpretation

Substituting Equation (54) into the full transformation gives

$$h'_\ell = \beta C_\ell h_\ell + \mu_\ell - \beta C_\ell \mu_\ell \tag{55}$$

$$= \mu_\ell + \beta C_\ell (h_\ell - \mu_\ell). \tag{56}$$

Therefore,

$$\boxed{h'_\ell = \mu_\ell + \beta C_\ell (h_\ell - \mu_\ell).} \tag{57}$$

This gives the clearest interpretation of the method:

1. subtract the activation centroid,
2. transform the centered activation using the conceptor,
3. restore the original centroid.

Thus, the method modifies the geometry of the activation *around* the centroid rather than moving the centroid itself.

1.4.3 Proof of mean preservation

Taking expectations of Equation (57),

$$\mathbb{E}[h'_\ell] = \mu_\ell + \beta C_\ell \mathbb{E}[h_\ell - \mu_\ell] \tag{58}$$

$$= \mu_\ell + \beta C_\ell \left(\mathbb{E}[h_\ell] - \mu_\ell\right) \tag{59}$$

$$= \mu_\ell. \tag{60}$$

Hence,

$$\boxed{\mathbb{E}[h'_\ell] = \mu_\ell.} \tag{61}$$

The mean-preservation property therefore follows directly from the choice of the affine bias and does not require an empirical approximation beyond the estimate of μ_ℓ.

1.4.4 Residual mean-corrected formulation

To combine mean preservation with controlled intervention, we introduce the residual coefficient λ:

$$\boxed{h'_\ell = h_\ell + \lambda \left[(\beta C_\ell - I) h_\ell + b_\ell \right],} \tag{62}$$

where

$$b_\ell = \mu_\ell - \beta C_\ell \mu_\ell.$$

Substituting the bias term,

$$\begin{aligned} h'_\ell &= h_\ell + \lambda \left[(\beta C_\ell - I) h_\ell + \mu_\ell - \beta C_\ell \mu_\ell \right] && (63) \\ &= h_\ell + \lambda (\beta C_\ell - I)(h_\ell - \mu_\ell). && (64) \end{aligned}$$

Therefore,

$$\boxed{h'_\ell = h_\ell + \lambda (\beta C_\ell - I)(h_\ell - \mu_\ell).} \tag{65}$$

At

$$\lambda = 0,$$

we recover

$$h'_\ell = h_\ell.$$

At

$$\lambda = 1,$$

we obtain

$$\begin{aligned} h'_\ell &= h_\ell + (\beta C_\ell - I)(h_\ell - \mu_\ell) && (66) \\ &= \mu_\ell + \beta C_\ell (h_\ell - \mu_\ell). && (67) \end{aligned}$$

Thus, the complete method interpolates continuously between the original representation and a mean-preserving conceptor transformation.

1.4.5 Proof of mean preservation under the residual formulation

Taking expectations of Equation (65),

$$\begin{aligned} \mathbb{E}[h'_\ell] &= \mathbb{E}[h_\ell] + \lambda (\beta C_\ell - I) \mathbb{E}[h_\ell - \mu_\ell] && (68) \\ &= \mu_\ell + \lambda (\beta C_\ell - I)(\mu_\ell - \mu_\ell) && (69) \\ &= \boxed{\mu_\ell}. && (70) \end{aligned}$$

Therefore, the residual formulation preserves the original activation mean for every

$$\lambda \in \mathbb{R},$$

provided that the bias is computed consistently from the same μ_ℓ.

In particular, for the intended range

$$\lambda \in [0, 1],$$

the method provides a continuous interpolation between no steering and full mean-preserving conceptor steering.

1.5 Unified Mathematical Interpretation

The four methods can now be interpreted as progressively richer models of the desired activation transformation.

Additive steering.

$$\boxed{h' = h + \beta v.} \tag{71}$$

The intervention is a translation in one learned direction. It is primarily a first-order, mean-based representation of the desired attribute.

Vanilla conceptor.

$$\boxed{h' = \beta C h.} \tag{72}$$

The intervention is covariance-aware and direction-dependent. The matrix C represents a soft subspace learned from the activation distribution.

Residual conceptor.

$$\boxed{h' = (1 - \lambda)h + \lambda \beta C h.} \tag{73}$$

The model retains a controlled fraction of the original activation, thereby introducing an explicit semantic-preservation parameter λ.

Mean-corrected residual conceptor.

$$\boxed{h' = h + \lambda(\beta C - I)(h - \mu).} \tag{74}$$

The transformation acts on deviations from the activation centroid, while the centroid itself is preserved.

The resulting hierarchy is therefore

$$\begin{array}{|rl|}\hline \text{Addition:} & h + \beta v, \\ \text{Conceptor:} & \beta C h, \\ \text{Residual Conceptor:} & h + \lambda(\beta C - I)h, \\ \text{Mean-Corrected Conceptor:} & h + \lambda(\beta C - I)(h - \mu). \\ \hline \end{array} \tag{75}$$

Each extension introduces one additional structural assumption:

Method	**Information used**	**Main motivation**
Addition	Mean difference	One dominant steering direction
Conceptor	Second-order geometry	Multi-directional structured behavior
Residual Conceptor	Geometry + residual	Preserve original representation
Mean-Corrected Conceptor	Geometry + residual + mean	Preserve centroid and semantics

1.6 Task Complexity and Expected Method Behavior

The above formulation also suggests a testable hypothesis concerning task complexity.

Suppose the behavior is effectively one-dimensional:

$$\mathcal{S} \approx \text{span}(v).$$

In this case, additive steering makes a minimal assumption and may be preferable because it modifies only the relevant direction.

By contrast, suppose the behavior occupies a structured k-dimensional subspace:

$$\mathcal{S} = \text{span}(v_1, \ldots, v_k), \qquad k > 1.$$

A single additive vector may be insufficient to represent the behavior. A conceptor can instead represent direction-dependent importance through

$$\gamma_i = \frac{\lambda_i}{\lambda_i + \alpha^{-2}}.$$

This leads to the following hypothesis:

$$\boxed{\text{Low effective dimensionality} \implies \text{additive steering may be sufficient,}} \tag{76}$$

while

$$\boxed{\text{high structured dimensionality} \implies \text{covariance-aware steering may become advantageous.}} \tag{77}$$

This is a hypothesis rather than a theorem: the actual performance depends on how the behavioral representation is encoded by the model.

A useful measure of the effective dimensionality of the activation distribution is the participation ratio

$$\boxed{r_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2},} \tag{78}$$

where λ_i are the eigenvalues of the activation correlation or covariance matrix.

If most of the variance is concentrated in a single direction, then

$$r_{\text{eff}} \approx 1.$$

If variance is distributed across many directions, then r_{eff} increases.

This provides an experimentally testable connection between the geometry of a task and the relative performance of additive versus conceptor-based steering.